

B.E. COMPUTER SCIENCE AND ENGINEERING
FOURTH YEAR FIRST SEMESTER - 2023

SUBJECT
PATTERN RECOGNITION

Time : 3 hours

Full marks : 100

Group A

Answer Question No. 1 (COMPULSORY) and any one form the rest of the two questions of this Group

1. Answer true or false stating reasons 2x10 = 20

- (a) A Naïve Bayes Classifier can only work on continuous-valued data.
- (b) The success of a pattern classification scheme using decision function depends only on finding the appropriate form of $d(x)$.
- (c) In a typical PR problem, the dimensionality of measurement space is always less than that of feature space.
- (d) Missing data in a database may be due to the error of the equipment in recording the data.
- (e) Data preprocessing is required to remove the noise if any from the data.
- (f) In Bayesian Network, the nodes represent events and the edges represent conditional dependencies.
- (g) For a supervised pattern classification problem having M classes, where the patterns are pair-wise separable, the classifier needs to compute $M/2$ number of decision surfaces to perform classification.
- (h) Data transformation methods provide a subset of the original data.
- (i) The merit of a NB classifier is that, the assumption of class conditional independence is always valid for real life problems.
- (j) Binning is a method of data preprocessing to detect and remove outliers.

2. (a) Explain briefly with an example the necessary steps involved in a typical PR system. 10

- (b) Assume any three disjoint pattern classes ω_1, ω_2 and ω_3 in $\mathbb{R}^2$. Compute the decision boundaries and decision rules that are needed to successfully classify all the patterns from the said three classes for the following two cases:

1. Each class is separable from all other classes
2. Classes are pair-wise separable

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3. (a) How binning and clustering can be used to handle noisy data? 5 + 5
- (b) Explain briefly a typical Naïve Bayes Classifier. What are the advantages of such a classifier over KNN and Decision Tree? 6 + 4

Group B

Answer Question No. 4 (COMPULSORY) and any two from the rest of the three questions of this Group

4. Answer true or false stating reasons 2x10 = 20

- (a) K-means algorithm cannot cluster data sets well with large differences in densities.
- (b) Each node in a typical ANN can have any number of outgoing connections, where the signals in all of these may not be the same.
- (c) A Syntactic Pattern Classifier infers a grammar from a set of training patterns.
- (d) Partitional clustering methods help in exploring data at different levels of granularity.
- (e) The support vectors of a linear SVM can be any points in the training set.
- (f) Syntactic Pattern Recognition attempts to classify patterns based on a set of extracted features and an underlying statistical model for the generation of these patterns.
- (g) While using RBF kernel with SVM, choosing a small value of 'Gamma' may overfit the classifier.
- (h) Hopfield network is usually employed for data clustering.
- (i) A multi-layer-perceptron is a recursive network that is suitable for solving optimization problem.
- (j) The training phase is not required in a typical Syntactic Pattern classification system.

5. What is partitional clustering? State and explain one such algorithm with example. Discuss the merits and demerits of the method. 2 + 12 + 6

6. (a) Explain the following terms in connection to a SVM Classifier 2x4

- (i) Maximum Margin Hyperplane
 - (ii) Kernel function
- (b) Suppose that two pattern classes are linearly separable. Describe a method using a suitable ANN that can successfully classify the said two patterns classes.

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7. Write short notes on any two of the following.

2 X 10

- (a) Importance of Syntactic PR
- (b) Evaluation of clustering results
- (c) Syntactic Pattern Description